Supporting Scientific Sovereignty: Evaluating the Feasibility of Partner-Led Respiratory Pathogen Surveillance in Global Health Alliances

Celia Bartels[1], Amy Ellis[1], Passwell Munachoonga[2], Mutinta Hamahuwa[2], Tom Aichele[1], Sam Baron[3], Catherine Sutcliffe[3], Shelby O'Connor[1], David O'Connor[1]

[1]UW-Madison School of Medicine and Public Health [2]Macha Research Trust [3]Johns Hopkins Bloomberg School of Public Health

INTRODUCTION

Figure 1: Causal Loop Diagram (CLD) of Sustainable Surveillance. This reinforcing loop illustrates how local data production builds scientific sovereignty and public trust, creating a "virtuous cycle" that justifies ongoing infrastructure investment and reduces dependence on extractive research models.

The Challenge: Global health surveillance historically relies on extractive research models, where data is collected locally but owned by external institutions, undermining local sustainability and scientific agency.

The Project: We co-created a decentralized, partner-led pathogen surveillance model in Choma, Zambia.

Research Question Can a decentralized, partner-led model maintain the same longitudinal data integrity and operational reliability as a directly managed domestic model?

Assessment: The sampling and testing data from the Zambia site was matched with a subset of data from a university-led site in Madison, Wisconsin, which served as a baseline for comparison. Data was normalized for the number of sampler machines, timeframe of data collection, and type of medical facility. Finally, result turnaround times and number and consistency of testing was compared between the two sites.

Results: Using the Cepheid GeneXpert to monitor common respiratory pathogens, the Zambia-led site achieved testing consistency and turnaround times comparable to, and often better than, the university-led site.

Conclusion: By prioritizing scientific sovereignty, we co-created an independent "node" for long-term management of endemic pathogens. This horizontal partnership serves as a blueprint for non-extractive global health infrastructure.

METHODS

Initiating an International Pathogen Surveillance Node:

Theory of Change Model

Goal: Apply a new model to global health epidemiological studies by creating sustainable international collaborations and supporting locally-managed environmental surveillance, intentionally rejecting extractive research.

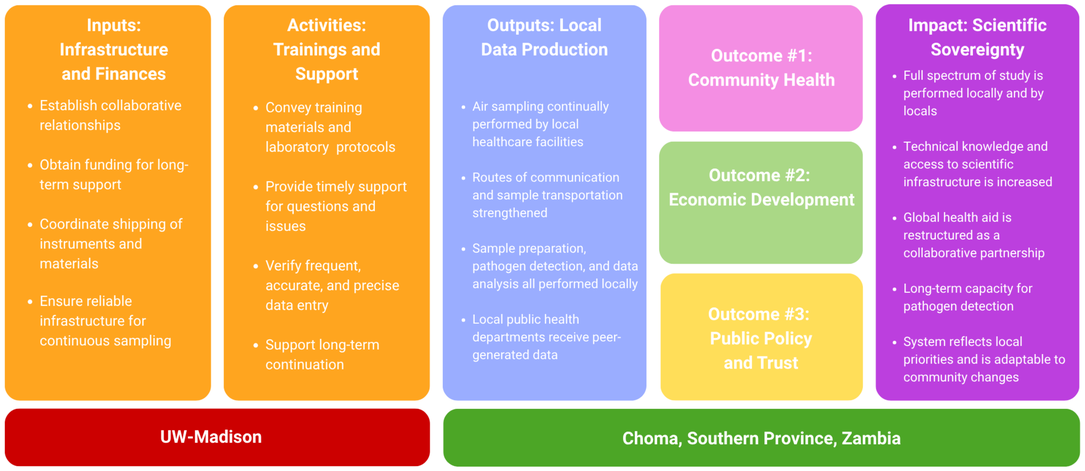

Figure 2: Theory of Change (ToC) for the Choma, Zambia pathogen surveillance site. This logic model maps the transition from initial inputs (equipment and training) provided by UW-Madison through decentralized activities to long-term health and economic impacts. The later-term outcomes and impacts of community health, economic development, public policy and trust, and scientific sovereignty represent broader effects of pathogen surveillance unique to lower-resource settings, emphasizing the many benefits of a long-term, locally-run system.

RESULTS

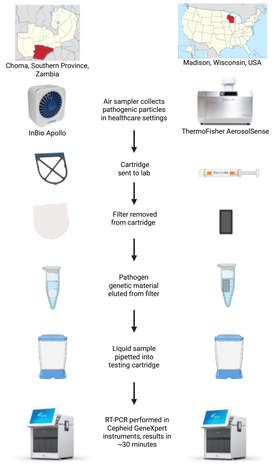

Figure 3: Air sampling and pathogen detection protocols developed for use in Choma, Zambia, and Madison, Wisconsin.

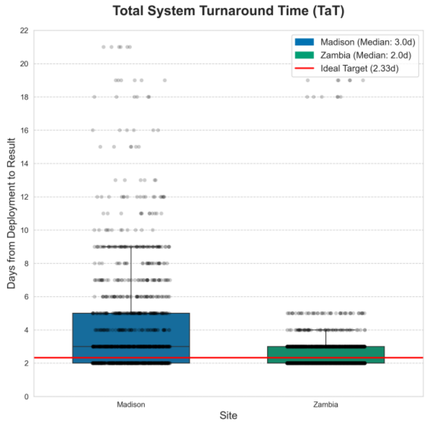

Figure 4: Operational efficiency from sampling start to PCR result. Boxplots represent the interquartile range (IQR) with internal lines indicating the median (Zambia: 2.0 days; Madison: 3.0 days); individual data points are shown via jittered overlay. The theoretical ideal for a 3x/week swap schedule is 2.33 days (red dashed line), the Zambian site excelled compared to the Madison site. A two-sample t-test ($p < 0.001$) confirms statistical significance of this difference.

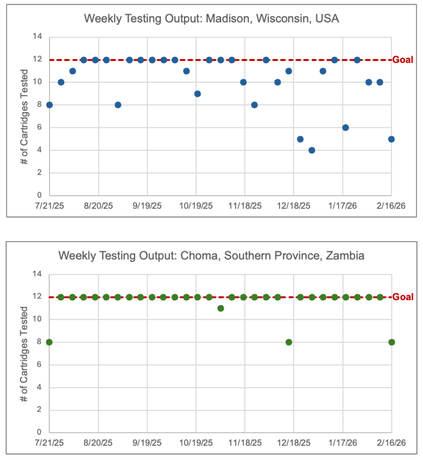

Figure 5: Weekly counts of cartridges tested at each surveillance site (July '25 - Feb '26). Both sites expected to test 12 samples/week. The Zambian site was more successful at reaching this goal (87.5% of weeks) compared to the Madison site (46.9% of weeks).

DISCUSSION

Significant variance in TaT (Fig. 4) and higher testing consistency (Fig. 5) reflects technical rigor at the Choma, Zambia site. While Madison faced challenges with healthcare staffing and competing priorities leading to collection lag, the Choma site maintained a 2.0-day median TaT, meeting operational targets despite infrastructure hurdles (e.g., insect ingress).

CONCLUSION

Our results confirm that decentralized, partner-led models not only maintain diagnostic integrity but can exceed the operational efficiency of domestic sites. The Madison-Choma partnership serves as a functional model for rejecting extractive research in favor of equitable, horizontal alliances—providing an essential foundation for global health equity.

ACKNOWLEDGEMENTS

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Data analysis and visualization were performed using Python (Matplotlib/Seaborn) and Microsoft Excel. Figures 1 and 2 were created in Canva. Figure 3 and overall poster layout were designed in BioRender.

Created with BioRender Poster Builder